

FP2 Chapter 11 Integration

Lecture 11 Integration

(Hyperbolic functions, Substitutions, Reduction formulae, Rectangle approximation, Arc Length, Surface Area of Revolution)

1. Important Results

- $\int f'(x)e^{f(x)}dx = e^{f(x)} + c$

Proof:

$$\frac{d}{dx}(e^{f(x)}) = e^{f(x)} \times f'(x)$$

Conversely, $\int f'(x)e^{f(x)}dx = e^{f(x)} + c$

Example 1:

(i) $\int xe^{x^2}dx$

(ii) $\int x^2e^{-x^3}dx$

(iii) $\int \cos x e^{\sin x}dx$

(iv) $\int \frac{e^{\sqrt{x}}}{\sqrt{x}}dx$

$$- \int f'(x) (f(x))^n dx = \frac{(f(x))^{n+1}}{n+1} + c, n \neq -1$$

$$\text{If } n = -1, \text{ then } \int f'(x) (f(x))^{-1} dx = \int \frac{f'(x)}{f(x)} dx = \ln|f(x)| + c$$

Proof:

$$\frac{d}{dx} \left(\frac{(f(x))^{n+1}}{n+1} \right) = (f(x))^n f'(x)$$

$$\text{Conversely, } \int f'(x) (f(x))^n dx = \frac{(f(x))^{n+1}}{n+1} + c, n \neq -1$$

Example 2:

$$(i) \quad \int \cos x \sin x dx$$

$$(ii) \quad \int x \sqrt{1+x^2} dx$$

$$(iii) \quad \int \frac{\ln x}{x} dx$$

$$(iv) \quad \int \frac{x}{\sqrt{1-x^2}} dx$$

2. Integration of Hyperbolic Functions

$$(i) \quad \frac{d}{dx}(\sinh mx) = \cosh mx \times m \Rightarrow \int \cosh mx \, dx = \frac{\sinh mx}{m} + c$$

$$(ii) \quad \frac{d}{dx}(\cosh mx) = \sinh mx \times m \Rightarrow \int \sinh mx \, dx = \frac{\cosh mx}{m} + c$$

$$(iii) \quad \frac{d}{dx}(\tanh mx) = \operatorname{sech}^2 mx \times m \Rightarrow \int \operatorname{sech}^2 mx \, dx = \frac{\tanh mx}{m} + c$$

- Most of the integrals involving hyperbolic functions can be evaluated by methods similar to those used for the corresponding trigonometric integrals (already done in P3).
- Some hyperbolic integrals, however, are best evaluated by replacing the hyperbolic functions by exponential functions.
- **Example 3: Integrate the following functions.**

$$(i) \quad \int \sinh^3 x \, dx$$

$$(ii) \quad \int \tanh x \, dx$$

$$(iii) \quad \int \coth x \, dx$$



(iv) $\int \sinh^2 x \, dx$

(v) $\int \cosh^2 2x \, dx$

(vi) $\int \tanh^2 x \, dx$

(vii) $\int \operatorname{sech} x \, dx$

(viii) $\int \operatorname{cosech} 2x \, dx$

Sunway A Levels

3. Integrals involving the function $\frac{1}{\sqrt{ax^2+bx+c}}$

- If $b = 0$,

Examples:

(i) $\int \frac{1}{\sqrt{4-x^2}} dx$

(ii) $\int \frac{1}{\sqrt{1+4x^2}} dx$

(iii) $\int \frac{1}{\sqrt{2x^2-1}} dx$



- If $b \neq 0$, by **completing the square** for the quadratic function $ax^2 + bx + c$, $\int \frac{1}{\sqrt{(ax^2+bx+c)}} dx$ can be expressed in one of the three following cases:

$$k \int \frac{1}{\sqrt{\{(x \pm p)^2 - q^2\}}} dx = k \cosh^{-1} \left(\frac{x \pm p}{q} \right) + c$$

$$k \int \frac{1}{\sqrt{\{(x \pm p)^2 + q^2\}}} dx = k \sinh^{-1} \left(\frac{x \pm p}{q} \right) + c$$

$$k \int \frac{1}{\sqrt{\{q^2 - (x \pm p)^2\}}} dx = k \sin^{-1} \left(\frac{x \pm p}{q} \right) + c$$

where k, p and q are constants.

- **Example 4:**

(i) $\int \frac{1}{\sqrt{2x-x^2}} dx$

(ii) $\int \frac{1}{\sqrt{x^2-6x+5}} dx$

(iii) $\int \frac{1}{\sqrt{x^2-6x+10}} dx$

(iv) Find $\int_{-1}^4 \frac{1}{\sqrt{4x^2 - 12x + 34}} dx$, giving your answer in the form $p \ln(q + r\sqrt{2})$.

(v) Find $\int_3^5 \frac{1}{\sqrt{(15+2x-x^2)}} dx$, giving your answer as a multiple of π .

Sunway A Levels

4. Integration requires the use of the substitution $t = \tan \frac{1}{2}x$.

$$t = \tan \frac{1}{2}x$$

$$\frac{dt}{dx} = \frac{1}{2} \sec^2 \frac{1}{2}x$$

$$\frac{dt}{dx} = \frac{1}{2} \left(1 + \tan^2 \frac{1}{2}x \right) = \frac{1}{2} (1 + t^2)$$

$$dx = \frac{2}{1 + t^2} dt$$

From MF19,

$$\sin x = \frac{2t}{1 + t^2} \quad \text{and} \quad \cos x = \frac{1 - t^2}{1 + t^2}$$

- **Example 5:**

- (i) Use the substitution $t = \tan \frac{1}{2}x$ to show that

$$\int_0^{\frac{\pi}{2}} \frac{1}{2 + \sin x} dx = \frac{\sqrt{3}}{9} \pi$$

- (ii) Find $\int \frac{1}{3 - \cos x} dx$.

5. Reduction Formulae

- Certain methods of integration which are applicable to functions involving a power n which n is a positive integer, are viable only when n is relatively small.
- For instance, $\int \cos^n x \, dx$ can be found when $n = 2$ or 4 by using the identity $\cos^2 A \equiv \frac{1}{2}(1 + \cos 2A)$ as often as necessary, but the same method applied to $\int \cos^{10} x \, dx$ would be extremely unwieldy.
- In cases like this, a means of symmetrically reducing the value of n is useful, and is called a **reduction method**.
- By far the most common method of establishing reduction formulae is by integration by parts. But it should not be assumed that this is the only approach.

- (l) The reduction formula can be obtained directly after applying integration by parts

- Case 1: $\int \sin^n x \, dx$, $\int \cos^n x \, dx$, $\int \sinh^n x \, dx$, $\int \cosh^n x \, dx$

Example 6:

Given that $I_n = \int_0^{\frac{\pi}{2}} \sin^n x \, dx$, where n is a non-negative integer. Show that

$$I_{n+2} = \frac{n+1}{n+2} I_n.$$

Hence, or otherwise, find I_5 .

Example 7:

$$I_n = \int \cosh^n x \, dx, n \geq 0$$

(a) Show that, for $n \geq 2$, $nI_n = \sinh x \cosh^{n-1} x + (n-1)I_{n-2}$.

(b) Hence find the exact value of $\int_0^{\ln 2} \cosh^5 x \, dx$.

- **Case 2:** $\int \tan^n x \, dx, \int \cot^n x \, dx, \int \tanh^n x \, dx, \int \coth^n x \, dx$

Example 8:

Let $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot^n x \, dx$, where $n \geq 0$. Show that, for $n \geq 2$, $I_n = \frac{1}{n-1} - I_{n-2}$.

Sunway A Levels

- **Example 9:**

$$I_n = \int_0^{\ln 2} \tanh^n x \, dx.$$

(a) Show that $I_n = I_{n-2} - \frac{1}{n-1} \left(\frac{3}{5}\right)^{n-1}$, $n \geq 2$.

(b) Hence find I_5 .

- **Case 3:** $\int \sec^n x \, dx, \int \operatorname{cosec}^n x \, dx, \int \operatorname{sech}^n x \, dx, \int \operatorname{cosech}^n x \, dx$

- **Example 10:**

Given that $I_n = \int \operatorname{cosec}^n x \, dx$, show that

$$(n-1)I_n = -\cot x \operatorname{cosec}^{n-2} x + (n-2)I_{n-2}.$$

Hence find the exact value of $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \operatorname{cosec}^6 x \, dx$.

- **Example 11:**

$$I_n = \int_0^{\ln 2} \operatorname{sech}^n x \, dx.$$

Show that $(n-1)I_n = (n-2)I_{n-2} + \left(\frac{3}{5}\right)\left(\frac{4}{5}\right)^{n-2}$.

[You may need $\frac{d}{dx}(\operatorname{sech} x) = -\tanh x \operatorname{sech} x$]



- **Case 4: Others**
- **Example 12:**

If $I_n = \int_0^1 \frac{x^n}{\sqrt{(x^2-1)}} dx$, show that $I_n = \left(\frac{n-1}{n}\right) I_{n-2}$.

Sunway A Levels

- **Example 13:**

Given that $I_n = \int_0^1 (1 - x^2)^n dx, n \geq 0$. Show that $I_n = \frac{2n}{2n+1} I_{n-1}, n \geq 1$.

Hence evaluate I_5 , leaving your answer as a fraction in its lowest terms.

Sunway A Levels

- **Example 14:**

If $I_n = \int_0^1 x^n e^{-x^2} dx$, prove that $2I_n = (n-1)I_{n-2} - e^{-1}, n \geq 2$.

Hence, find I_5 , leaving your answer in terms of e .

Sunway A Levels

(II) The reduction formula can be obtained by applying integration by parts twice

- **Example 15:**

Given that $I_n = \int_0^{\frac{\pi}{2}} e^{-x} \sin^n x \, dx$, show that $(n^2 + 1)I_n = -e^{-\frac{\pi}{2}} + n(n - 1)I_{n-2}$, $n \geq 2$. Hence show that $I_4 = \frac{1}{85} \left(24 - 41e^{-\frac{\pi}{2}} \right)$.

- **Example 16:**

Given that $I_n = \int_0^1 (1-x)^n \sinh x \, dx$, show that, for $n \geq 2$, $I_n = -1 + n(n-1)I_{n-2}$. Hence find the exact value of I_4 .

Sunway A Levels

(III) Other methods of establishing reduction formulae**- Example 17:**

It is given that $I_n = \int_0^1 x^n e^{-x^3} dx, n \geq 0$. By considering $\frac{d}{dx}(x^{n+1}e^{-x^3})$, show that

$I_{n+3} = \frac{1}{3}(n+1)I_n - \frac{1}{3}e^{-1}$. Hence evaluate I_8 , leaving your answer in terms of e .

- **Example 18:**

Given that $I_n = \int_1^e (1 + \ln x)^n dx$. By considering $\frac{d}{dx}(x(1 + \ln x)^{n+1})$ or otherwise, show that $I_{n+1} = (2^{n+1})e - 1 - (n + 1)I_n$.

Sunway A Levels

- **Example 19:**

Given that $I_n = \int_0^1 (1+x^2)^{-n} dx, n \geq 1$, show by considering $\frac{d}{dx}\{x(1+x^2)^{-n}\}$, or otherwise, that $2nI_{n+1} = 2^{-n} + (2n-1)I_n$.

Hence, or otherwise, find the value of I_3 , leaving your answer in terms of π .

Sunway A Levels



6. Rectangle Approximation to the Area under the Curve

- Note that the definite integral $\int_a^b f(x) dx$ is an area bounded by the curve $y = f(x)$, the x -axis, and the lines $x = a$ and $x = b$.

- If we divide the interval $a \leq x \leq b$ into n equal strips, the x -coordinates of the strips are $a, a + \frac{b-a}{n}, a + 2\left(\frac{b-a}{n}\right), a + 3\left(\frac{b-a}{n}\right), \dots b$.

- Hence the sum of the areas of all n rectangles **inside** the integral, shown on Figure A, is

$$\left\{ f(a) + f\left(a + \frac{b-a}{n}\right) + f\left(a + 2\frac{b-a}{n}\right) + \dots + f\left(a + (n-1)\frac{b-a}{n}\right) \right\} \left(\frac{b-a}{n}\right)$$

and the sum of the areas of all n rectangles **outside** the integral, shown in Figure B, is

$$\left\{ f\left(a + \frac{b-a}{n}\right) + f\left(a + 2\frac{b-a}{n}\right) + \dots + f\left(a + (n-1)\frac{b-a}{n}\right) + f(b) \right\} \left(\frac{b-a}{n}\right)$$



- The actual value of the definite integral is between these two values. As n tends to infinity, these two sums tends to the same value.
- Hence, we have

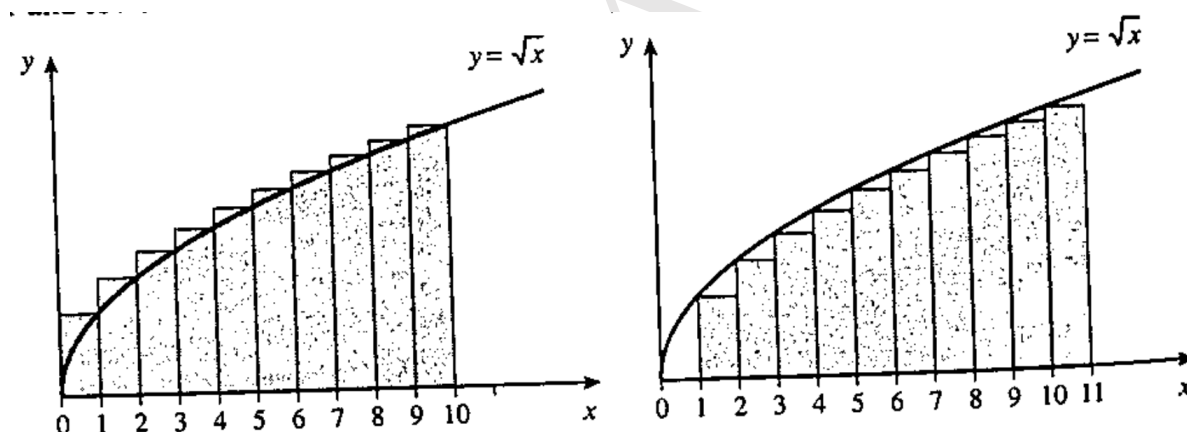
$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \left\{ f(a) + f\left(a + \frac{b-a}{n}\right) + f\left(a + 2\frac{b-a}{n}\right) + \dots + f\left(a + (n-1)\frac{b-a}{n}\right) \right\} \left(\frac{b-a}{n}\right)$$

which gives

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{r=0}^{n-1} f\left(a + r\frac{b-a}{n}\right) \left(\frac{b-a}{n}\right)$$

- That is, the integral is the limit as n tends to infinity of the sum of the series. This method may be used to find upper and lower bounds of integrals and series.
- **Worked Example 1:**

Find upper and lower bounds for $\sqrt{1} + \sqrt{2} + \sqrt{3} + \dots + \sqrt{10}$.



- In the left diagram of Figure the sum is represented as the area of a set of 10 rectangles, each having width 1 and with heights $\sqrt{1}, \sqrt{2}, \sqrt{3}, \dots, \sqrt{10}$. These cover the interval $0 \leq x \leq 10$ of the x -axis.
- The top right corners of these rectangles have coordinates $(1, \sqrt{1}), (2, \sqrt{2}), (3, \sqrt{3}), \dots, (10, \sqrt{10})$ all of which lie on the curve $y = \sqrt{x}$.
- Clearly the sum of the areas of the rectangles is greater than the area under the curve, so that



$$\sum_{n=1}^{10} \sqrt{n} > \int_0^{10} \sqrt{x} dx = \left[\frac{2}{3} x^{\frac{3}{2}} \right]_0^{10} = \frac{20}{3} \sqrt{10}$$

gives a **lower bound** for the sum.

- To find an upper bound, push the all rectangles to the right by 1 unit, so that they cover the interval $1 \leq x \leq 11$. The right diagram of Figure shows that the top left corners of the rectangles $y = \sqrt{x}$, and it is easy to see that

$$\sum_{n=1}^{10} \sqrt{n} < \int_1^{11} \sqrt{x} dx = \left[\frac{2}{3} x^{\frac{3}{2}} \right]_1^{11} = \frac{22}{3} \sqrt{11} - \frac{2}{3}$$

gives a **upper bound** for the sum.

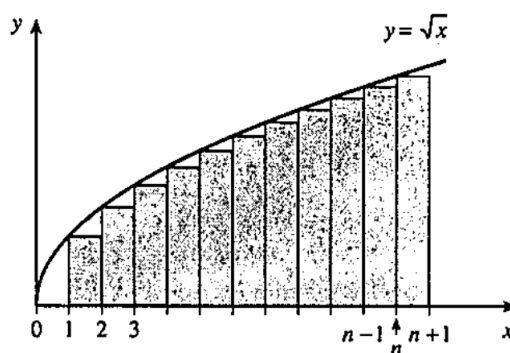
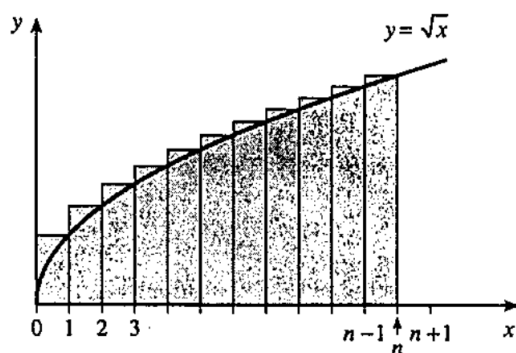
- So,

$$\frac{20}{3} \sqrt{10} < \sum_{n=1}^{10} \sqrt{n} < \frac{22}{3} \sqrt{11} - \frac{2}{3}$$



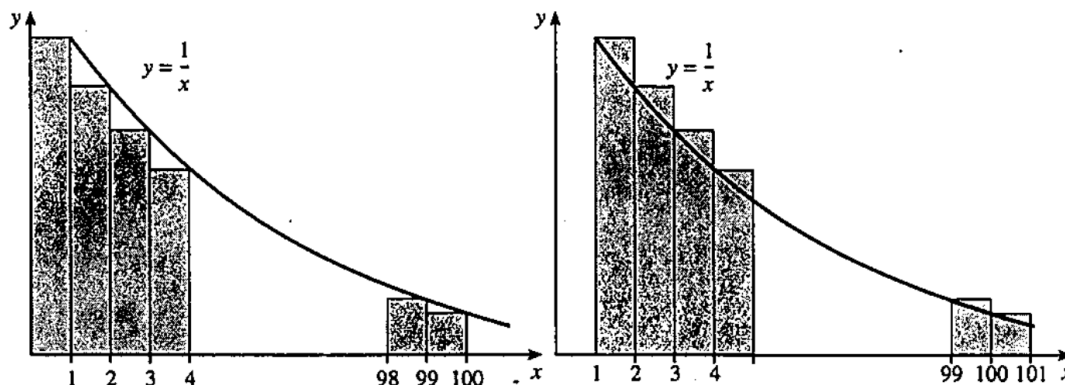
- If we extend this idea to find the sum of the sum of the first n square roots, can you find the lower bound and upper bound?

Find lower and upper bounds for the sum $\sqrt{1} + \sqrt{2} + \sqrt{3} + \dots + \sqrt{n}$.



- **Worked Example 2:**

Show that $\ln 101 < 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{100} < 1 + \ln 100$.



- The right diagram of the figure shows the sum $1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{100}$ represented as a set of 100 rectangles, each having width 1 and heights $1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{100}$. These cover the interval $1 \leq x \leq 101$ of the x -axis. The top left corners of these rectangles have coordinates $(1,1), (2, \frac{1}{2}), (3, \frac{1}{3}), \dots, (100, \frac{1}{100})$ all of which lie on the curve $y = \frac{1}{x}$. Clearly the sum of areas of the rectangles is greater than area of the curve, so that

$$1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{100} > \int_1^{101} \frac{1}{x} dx = \ln 101$$

- The left diagram shows all the shaded rectangle having moved to the left by 1 unit. However, there is a problem, because you cannot have 0 as the lower limit of this integral which would then not exist. The solution is to ignore the first rectangle: the area under the curve between 1 and 100 is greater than the sum of the areas of the remaining rectangles, so

$$\frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{100} < \int_1^{100} \frac{1}{x} dx = \ln 100$$

- Therefore, adding 1 (the area of the first rectangle) to both sides,

$$1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{100} < 1 + \ln 100$$

- So, $\ln 100 < 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{100} < 1 + \ln 101$

- **Example 20:**

The diagram shows the parts of the graph of $y = \frac{1}{1+x}$ between $x = 0$ and $x = 1$.

The four rectangles drawn under the curve are of equal width, and their total area is an approximate to the area under the curve from $x = 0$ and $x = 1$. Calculate the approximation to the area under the curve, giving 3 significant figures in your answer.



When there are n rectangles of equal width under the curve between $x = 0$ and $x = 1$ (instead of just four), find an expression for their total area, and show that it may be written as

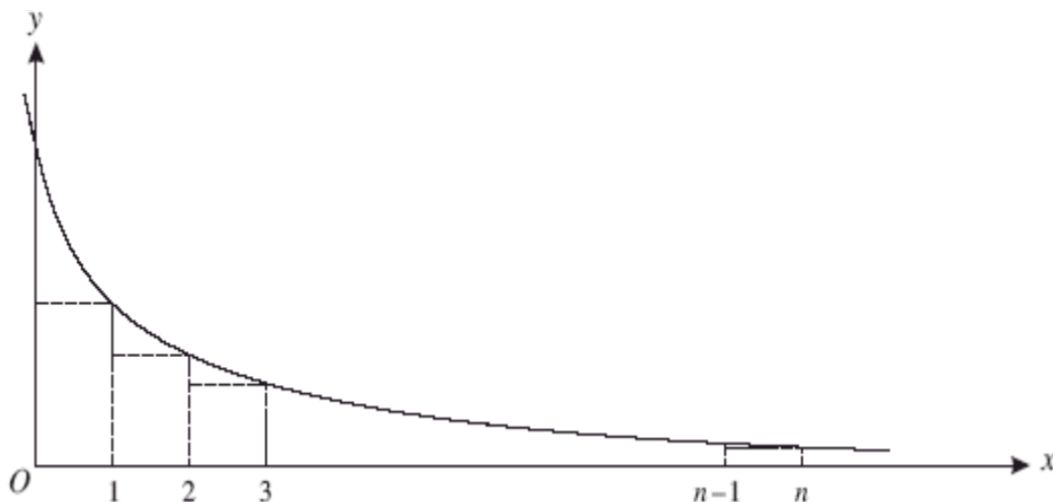
$$\sum_{r=1}^n \frac{1}{n+r}.$$

Deduce that

$$\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{n+r} = \ln 2.$$



- **Example 21:**



The diagram shows the curve with equation $y = \frac{1}{x+1}$. A set of n rectangles of unit width is drawn, starting at $x = 0$ and ending at $x = n$, where n is an integer.

- (i) By considering the areas of these rectangles, explain why

$$\frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n+1} < \ln(n+1).$$

- (ii) By considering the areas of another set of rectangles, show that

$$1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n} > \ln(n+1).$$

- (iii) Hence show that

$$\ln(n+1) + \frac{1}{n+1} < \sum_{r=1}^{n+1} \frac{1}{r} < \ln(n+1) + 1.$$

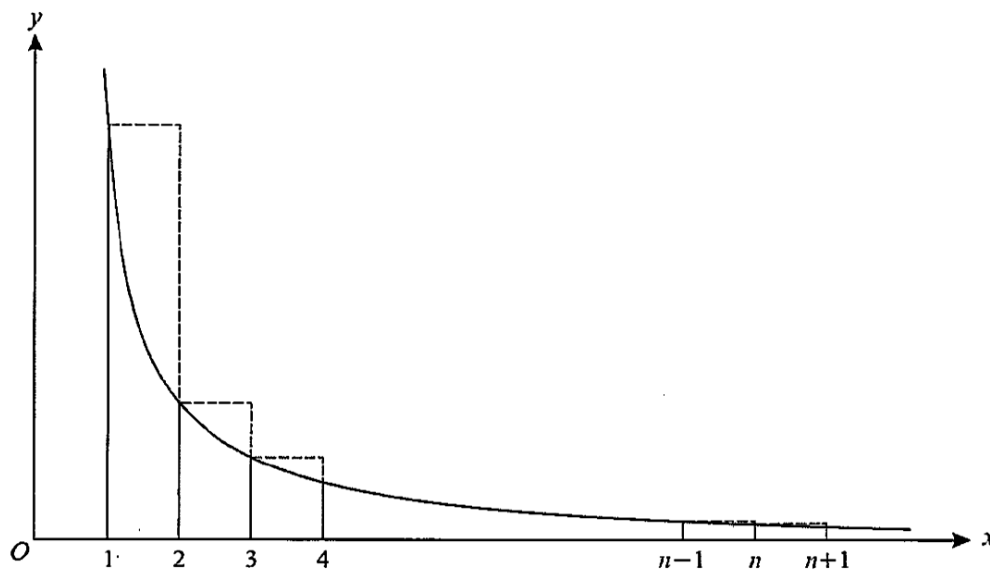
- (iv) State, with a reason, whether $\sum_{r=1}^{\infty} \frac{1}{r}$ is convergent.



Sunway A Levels



- **Example 22:**



The diagram shows the curve with equation $y = \frac{1}{x^2}$ for $x > 0$, together with a set of n rectangles of unit width, starting at $x = 1$.

(i) By considering the areas of these rectangles, explain why

$$\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{n^2} > \int_1^{n+1} \frac{1}{x^2} dx.$$

(ii) By considering the areas of another set of rectangles, explain why

$$\frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots + \frac{1}{n^2} < \int_1^n \frac{1}{x^2} dx.$$

(iii) Hence show that

$$1 - \frac{1}{n+1} < \sum_{r=1}^n \frac{1}{r^2} < 2 - \frac{1}{n}.$$

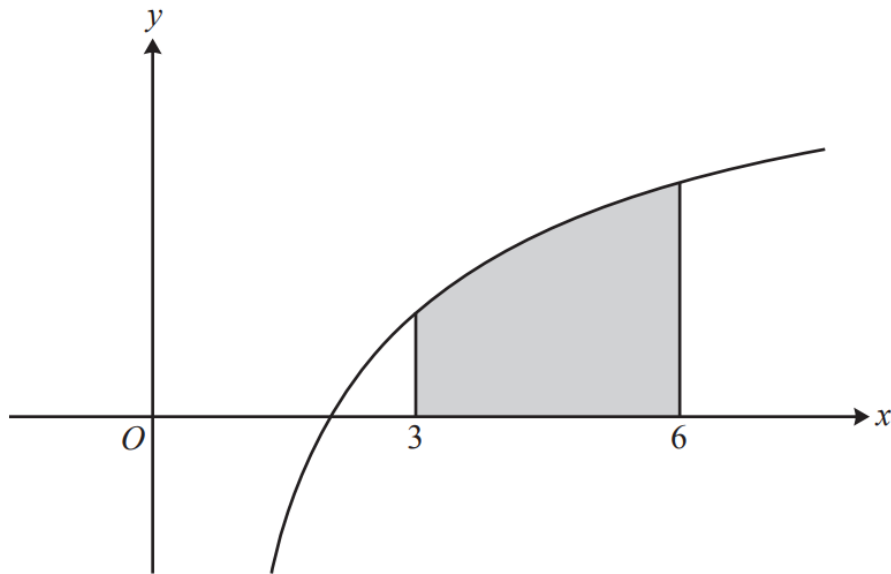
(iv) Hence give bounds between which $\sum_{r=1}^{\infty} \frac{1}{r^2}$ lies.



Sunway A Levels



- **Example 23:**



The diagram shows part of the curve $y = \ln(\ln x)$. The region between the curve and the x -axis for $3 \leq x \leq 6$ is shaded.

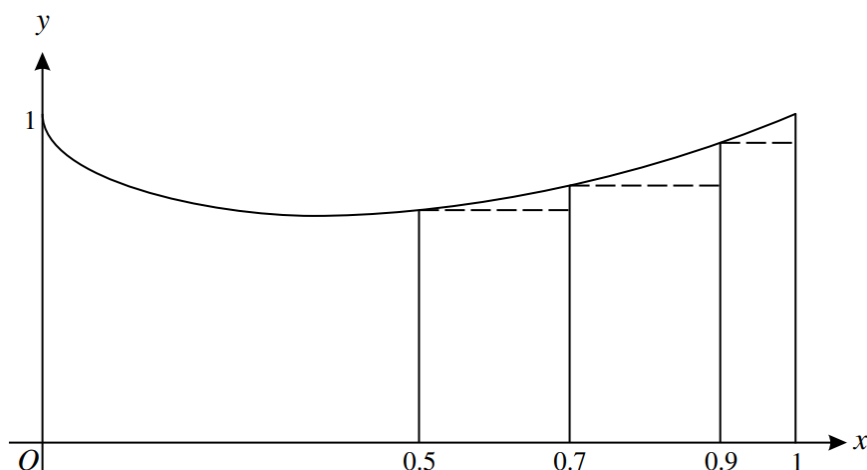
- (i) By considering n rectangles of equal width, show that a lower bound, L , for the area of the bounded region is $\frac{3}{n} \sum_{r=0}^{n-1} \ln \left(\ln \left(3 + \frac{3r}{n} \right) \right)$.
- (ii) By considering another set of n rectangles of equal width, find a similar expression for an upper bound, U , for the area of shaded region.
- (iii) Find the least value of n for which $U - L < 0.001$.



Sunway A Levels



- **Example 24:**



The diagram shows the curve $y = f(x)$, defined by

$$f(x) = \begin{cases} x^x & \text{for } 0 < x \leq 1 \\ 1 & \text{for } x = 0 \end{cases}$$

- (i) By first taking logarithms, show that the curve has a stationary point at $x = e^{-1}$.

The area under the curve from $x = 0.5$ to $x = 1$ is denoted by A .

- (ii) By considering the set of three rectangles shown in the diagram, show that a lower bound for A is 0.388.
- (iii) By considering another set of three rectangles, find an upper bound for A , giving 3 decimal places in your answer.

The area under the curve from $x = 0$ to $x = 0.5$ is denoted by B .

- (iv) Draw a diagram to show rectangles which could be used to find lower and upper bounds for B , using not more than three rectangles for each bound. (You are not required to find the bounds.)

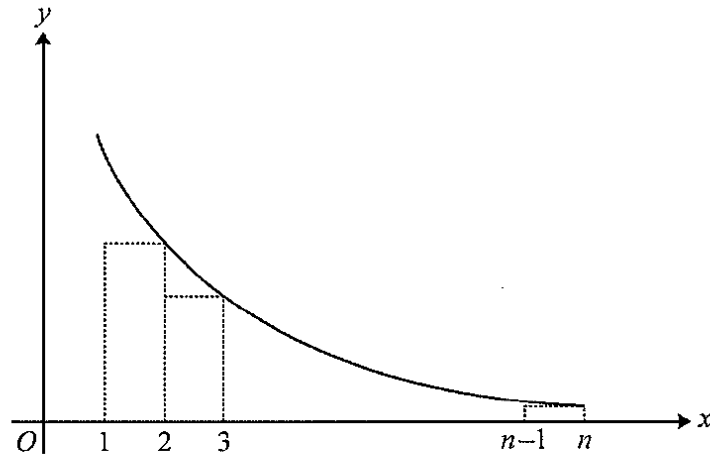


Sunway A Levels



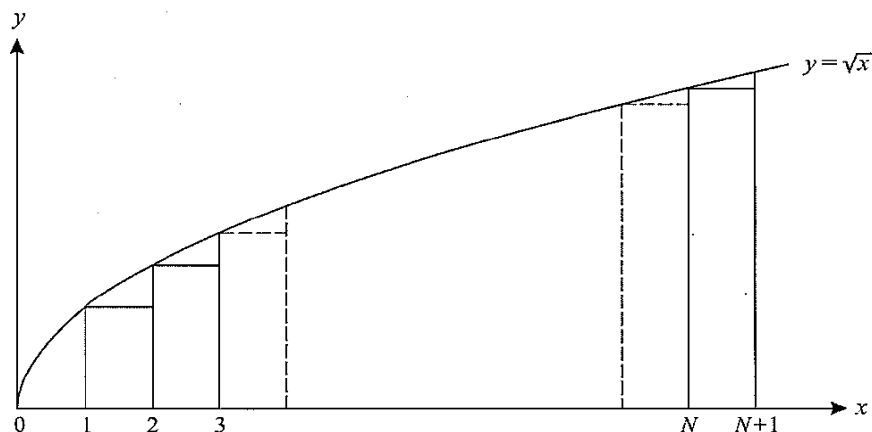
- **Exercise:**

The diagram shows the curve $y = \frac{1}{x^3}$ for $1 \leq x \leq n$ where n is an integer. A set of $(n-1)$ rectangles of width 1 is drawn under the curve.



(i) Write down the sum of the areas of the rectangles.

(ii) Hence show that $\sum_{r=1}^{\infty} \frac{1}{r^3} < \frac{3}{2}$.



The diagram shows the curve with equation $y = \sqrt{x}$. A set of N rectangles of unit width is drawn, starting at $x = 1$ and ending at $x = N + 1$, where N is an integer (see diagram).

- (i) By considering the areas of these rectangles, explain why

$$\sqrt{1} + \sqrt{2} + \sqrt{3} + \dots + \sqrt{N} < \int_1^{N+1} \sqrt{x} \, dx.$$

- (ii) By considering the areas of another set of rectangles, explain why

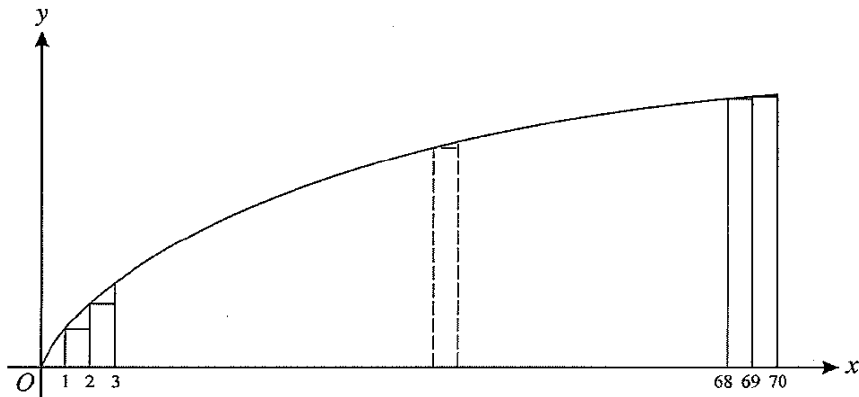
$$\sqrt{1} + \sqrt{2} + \sqrt{3} + \dots + \sqrt{N} > \int_0^N \sqrt{x} \, dx.$$

- (iii) Hence find, in terms of N , limits between which $\sum_{r=1}^N \sqrt{r}$ lies.



(i) Prove that $\int_0^N \ln(1+x) \, dx = (N+1) \ln(N+1) - N$, where N is a positive constant.

(ii)



The diagram shows the curve $y = \ln(1+x)$, for $0 \leq x \leq 70$, together with a set of rectangles of unit width.

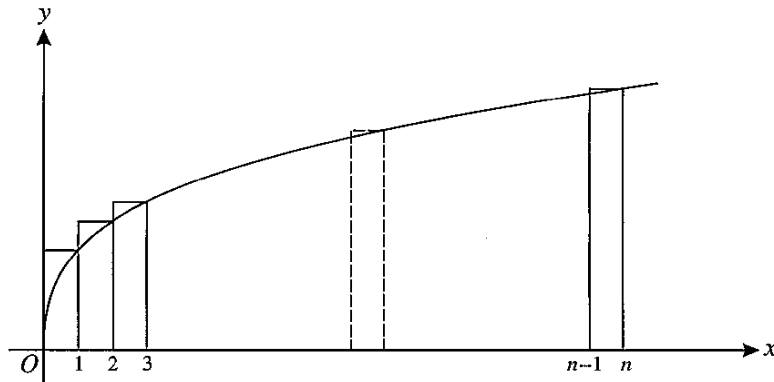
(a) By considering the areas of these rectangles, explain why

$$\ln 2 + \ln 3 + \ln 4 + \dots + \ln 70 < \int_0^{70} \ln(1+x) \, dx.$$

(b) By considering the areas of another set of rectangles, show that

$$\ln 2 + \ln 3 + \ln 4 + \dots + \ln 70 > \int_0^{69} \ln(1+x) \, dx.$$

(c) Hence find bounds between which $\ln(70!)$ lies. Give the answers correct to 1 decimal place.



The diagram shows the curve with equation $y = \sqrt[3]{x}$, together with a set of n rectangles of unit width.

- (i) By considering the areas of these rectangles, explain why

$$\sqrt[3]{1} + \sqrt[3]{2} + \sqrt[3]{3} + \dots + \sqrt[3]{n} > \int_0^n \sqrt[3]{x} \, dx.$$

- (ii) By drawing another set of rectangles and considering their areas, show that

$$\sqrt[3]{1} + \sqrt[3]{2} + \sqrt[3]{3} + \dots + \sqrt[3]{n} < \int_1^{n+1} \sqrt[3]{x} \, dx.$$

- (iii) Hence find an approximation to $\sum_{n=1}^{100} \sqrt[3]{n}$, giving your answer correct to 2 significant figures.

7. Arc Length of a Curve

- Our objective here is to find expressions for s , the arc length of a curve, in terms of the coordinates in which the equations are commonly written.

Suppose an arc of a curve is divided into n parts by the points $P_0, P_1, \dots, P_n$. We shall assume that the sum of the lengths of chords $\rightarrow$ length of the arc as the length of the chords $\rightarrow 0$.

Case 1: Cartesian Equations of Curves

$$s = \int_{x_1}^{x_2} \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

- **Example 25:**

The curve C is defined for $x \geq 0$ and has equation $y = \frac{1}{3}x^{\frac{3}{2}} - x^{\frac{1}{2}} + 1$. Find the length of the arc of C which joins $\left(1, \frac{1}{3}\right)$ to $\left(4, \frac{5}{3}\right)$.

Sunway A Levels

Case 2: Parametric Equations of Curves

$$s = \int_{t_1}^{t_2} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

- **Example 26:**

A curve is defined parametrically by $x = t^2 - 2 \ln t$, $y = 4(t - 1)$, $t \geq 1$. The points A and B on the curve are given by $t = 1$ and $t = 2$ respectively. Show that the length of the arc AB of the curve is $3 + 2 \ln 2$.

- **Example 27:**

Let $I_n = \int_0^1 t^n e^{-t} dt, n \geq 0$. Prove that for $n \geq 1, I_n = -e^{-1} + nI_{n-1}$.

For a general (x, y) on a curve C it is known that $\frac{dx}{dt} = t^4 e^{-t} \cos t$ and $\frac{dy}{dt} = t^4 e^{-t} \sin t$ where t is a real parameter. Show that the length of that part of C from the point where $t = 0$ to the point where $t = 1$ is equal to $24 - 65e^{-1}$.

Sunway A Levels

- **Example 28:**

The curve C has equation $y = \ln\left(\tanh\frac{1}{2}x\right)$, where $x > 0$.

- (a) Show that $\frac{dy}{dx} = \operatorname{cosech} x$.
- (b) Show that the length of part C from the point $x = 1$ to the point where $x = 3$ is $\ln(1 + e^2 + e^4) - 2$.



Case 3: Polar Equation of the Curve

Sunway A Levels

$$s = \int_{\theta_1}^{\theta_2} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$

- **Example 29:**

The curve C has polar equation $r = e^{4\theta}$ for $0 \leq \theta \leq \alpha$, where α measured in radians. The length of C is 2015. Find the value of α .

Sunway A Levels

- **Example 30:**

The curve C has polar equation $r = ae^{\frac{\theta}{k}}$, where a and k are positive constants and $0 \leq \theta \leq \pi$. Show that the length of C is $a\sqrt{k^2 + 1} \left(e^{\frac{\pi}{k}} - 1 \right)$.

- **Example 31:**

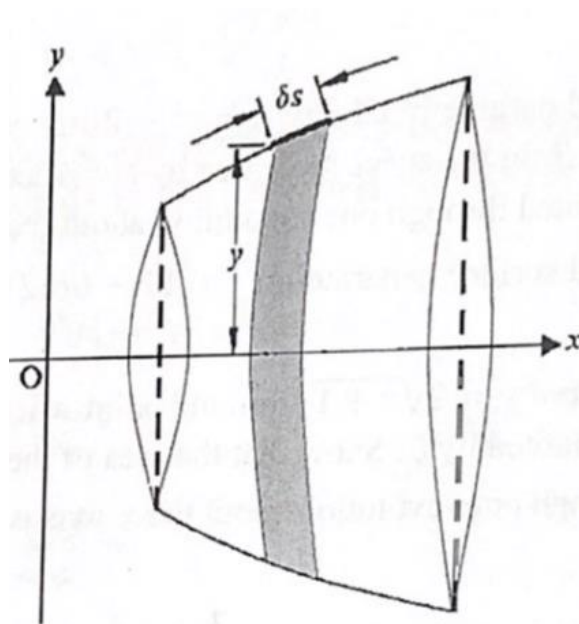
The area C has polar equation $r = a(1 + \sin \theta)$, where $0 \leq \theta \leq \frac{\pi}{2}$. Find the length of C .

- **Example 32:**

Verify that $\frac{d}{dx}(\sinh^{-1} x + x\sqrt{x^2 + 1}) = 2\sqrt{x^2 + 1}$.

Hence show that the length of the arc of the curve whose polar equation is $r = 3\theta$ from the point where $\theta = 0$ to the point where $\theta = 5$ is $\frac{3}{2}\ln(5 + \sqrt{26}) + \frac{15}{2}\sqrt{26}$.

8. Area of Surface of Revolution



- When a section of a curve rotates through one revolution about the x -axis, a three-dimensional surface of revolution is formed. The surface area of such an object can be found by summing the areas of small shells of the total surface.
- A typical shell is formed by the rotation of an arc of length δs tracing out a ring of radius y and width δs .
- The surface area, δS , of this ring is given by $\delta S \approx 2\pi y \delta s$.
- Now the equation of the curve whose rotation has formed the surface may be in Cartesian or parametric form (polar form is not in the syllabus).

$$\delta S \approx 2\pi y \delta s$$

$$\frac{\delta S}{\delta s} \approx 2\pi y, \text{ as } \delta s \rightarrow 0, \frac{\delta S}{\delta s} \rightarrow \frac{dS}{ds}$$

$$\frac{dS}{ds} = 2\pi y$$

Integrating both sides with respect to s ,

$$S = \int 2\pi y ds$$

Cartesian Equation

$$S = \int_{x_1}^{x_2} 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

- Example 33:

The arc of the curve $y = 2\sqrt{x+1}$ from the point where $x = 0$ to the point where $x = 1$ is denoted by C . Show that the area of the surface generated when C is rotated through one revolution about x -axis is $\frac{8}{3}\pi(3\sqrt{3} - 2\sqrt{2})$.

- **Example 34:**

The arc of the curve with equation $y = \frac{3}{8}x^{\frac{4}{3}} - \frac{3}{4}x^{\frac{2}{3}}$, from the point where $x = 1$ to the point where $x = 8$, is denoted by C . Find the area of the surface generated when C is rotated through one revolution about the x -axis.

Sunway A Levels

- **Example 35:**

A curve is defined by the equation $y = 2 + \cosh x$, where x is real. Calculate, correct to 3 significant figures, the area of the surface generated when the arc of the curve from the point where $x = 0$ to the point where $x = 3$ is rotated through one revolution about the x -axis.

Sunway A Levels

- **Example 36:**

The curve C has equation $y = e^{-x}$. The part of the curve between $x = 0$ and $x = \ln 3$ is rotated through 2π radians about the x -axis.

Show that the area S of the curved surface generated is given by

$$S = 2\pi \int_0^{\ln 3} e^{-x} \sqrt{1 + e^{-2x}} dx.$$

Use the substitution $e^{-x} = \sinh u$ to find the value of S , giving your answer to 3 decimal places.

Parametric Equation

$$S = \int_{t_1}^{t_2} 2\pi y \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

- Example 37:

A curve is defined parametrically by $x = t^2 - 2 \ln t$, $y = 4(t - 1)$, $t \geq 1$. The points A and B on the curve are given by $t = 1$ and $t = 2$ respectively. The arc AB is rotated through one revolution about the x -axis. Show that the area of the curved surface generated is $\frac{8}{3}\pi(11 - 6 \ln 2)$.